

Network Applications - Quiz Questions

Questions 1-20: Introduction to Network Applications and Key Concepts

1. What has been the driving force behind the Internet's success according to the lecture?

- A) High-speed hardware
- B) Numerous useful and entertaining applications ✓
- C) Government regulations
- D) Advanced encryption techniques

2. Which of the following is an example of a classic text-based Internet application from the 1970s and 1980s?

- A) Video conferencing
- B) Text e-mail ✓
- C) Social networking
- D) Mobile payment apps

3. What was the killer application of the mid-1990s mentioned in the lecture?

- A) E-mail
- B) File transfers
- C) The World Wide Web ✓
- D) Newsgroups

4. Which applications emerged in the new millennium as compelling Internet applications?

- A) Text-based e-mail only
- B) Voice over IP and video conferencing like Skype ✓
- C) Remote access to computers
- D) File transfers exclusively

5. What type of applications have created human networks on top of the Internet's network?

- A) File-sharing apps
- B) Social networking applications like Facebook and Twitter ✓
- C) E-commerce platforms
- D) Search engines

6. With the arrival of smartphones and 4G/5G, what profusion of apps has occurred?

- A) Desktop-only apps
- B) Location-based mobile apps like Yelp and Tinder ✓
- C) Text-only messaging
- D) Wired networking tools

7. In this chapter, what key concepts are defined for the application layer?

- A) Network services required by applications, clients and servers, processes, and transport-layer interfaces ✓
- B) Physical layer cabling standards
- C) Routing algorithms
- D) Encryption keys

8. Which network applications are examined in detail in the chapter?

- A) Web, e-mail, DNS, P2P file distribution, and video streaming ✓
- B) Operating systems and hardware drivers
- C) Database management systems
- D) Spreadsheet software

9. What is covered in the chapter regarding network application development?

- A) Development over TCP and UDP, including the socket interface ✓
- B) Hardware assembly
- C) Low-level bit manipulation
- D) Physical media types

10. How is an application's architecture different from the network architecture?

- A) Application architecture is designed by the developer and dictates structure over end systems ✓
- B) They are identical
- C) Network architecture is variable while application is fixed
- D) Application architecture focuses on routers

11. What are the two predominant architectural paradigms for modern network applications?

- A) Client-server and peer-to-peer (P2P) ✓
- B) Centralized and hierarchical
- C) Wired and wireless
- D) TCP and UDP

12. In client-server architecture, what is the role of the server?

- A) It is an always-on host that services requests from many clients ✓
- B) It initiates requests
- C) It is intermittently connected
- D) It only receives data

13. In the Web application, how do clients communicate in client-server architecture?

- A) Clients do not directly communicate with each other ✓
- B) Clients communicate directly via P2P
- C) Clients use routers as intermediaries
- D) Clients broadcast to all servers

14. What characteristic does the server have in client-server architecture?

- A) A fixed, well-known IP address ✓
- B) Variable addresses
- C) No address requirement
- D) Peer-generated addresses

15. Which applications use client-server architecture?

- A) Web, FTP, Telnet, and e-mail ✓
- B) Video games only
- C) Mobile apps exclusively
- D) Offline tools

16. Why might a single-server host be insufficient in client-server applications?

- A) It may be overwhelmed by requests ✓
- B) It has unlimited capacity
- C) It requires no data center
- D) It handles all requests instantly

17. What is often used to create a powerful virtual server in client-server?

- A) A data center with a large number of hosts ✓
- B) A single laptop
- C) Peer devices only
- D) Mobile phones

18. Which popular services run in data centers?

- A) Search engines, Internet commerce, Web-based e-mail, social media ✓

- B) Local file storage
- C) Hardware repairs
- D) Offline backups

19. In P2P architecture, what is the reliance on dedicated servers?

- A) Minimal or none ✓
- B) Complete reliance
- C) Partial for backups
- D) Only for authentication

20. What do peers do in P2P architecture?

- A) Communicate directly between intermittently connected hosts ✓
- B) Rely solely on servers
- C) Use centralized databases
- D) Avoid file sharing

Questions 21-40: P2P Architecture, Processes Communicating, and Client/Server Processes

21. Where do most peers reside in P2P?

- A) In homes, universities, and offices ✓
- B) In data centers only
- C) In servers exclusively
- D) In mobile networks

22. What is a compelling feature of P2P architectures?

- A) Self-scalability ✓
- B) High cost
- C) Centralized control
- D) Fixed bandwidth

23. In P2P file-sharing, how does self-scalability work?

- A) Each peer adds service capacity by distributing files ✓
- B) Peers reduce capacity
- C) Servers handle all load
- D) Bandwidth decreases

24. What challenges do P2P applications face?

- A) Security, performance, and reliability due to decentralization ✓
- B) Low scalability
- C) Over-reliance on servers
- D) Fixed structures

25. What is a process in the context of operating systems?

- A) A program running within an end system ✓
- B) A hardware component
- C) A network router
- D) A data packet

26. How do processes on the same end system communicate?

- A) With interprocess communication governed by the OS ✓
- B) Directly over the Internet
- C) Via routers
- D) Using physical media

27. In a network application, what consists of pairs of processes?

- A) Processes that send messages over a network ✓
- B) Isolated programs
- C) Hardware pairs
- D) Single processes only

28. In the Web application, what are the processes?

- A) Client browser and Web server ✓
- B) Two servers
- C) Two clients
- D) Routers

29. In P2P file-sharing, how is a peer labeled during file transfer?

- A) Downloading peer as client, uploading as server ✓
- B) Both as servers
- C) Both as clients
- D) No labels

30. Can a process in P2P file-sharing be both client and server?

- A) Yes, it can upload and download files ✓

- B) No, only one role
- C) Only in client-server
- D) Only for Web

31. How is the client defined in a communication session?

- A) The process that initiates the communication ✓
- B) The process that waits
- C) The always-on host
- D) The receiver only

32. What is the server in a communication session?

- A) The process that waits to be contacted ✓
- B) The initiator
- C) The mobile process
- D) The intermittent one

33. In the Web, which is the client?

- A) The browser process ✓
- B) The Web server
- C) The router
- D) The DNS

34. In P2P file sharing, when Peer A asks Peer B for a file, what roles do they play?

- A) Peer A is client, Peer B is server ✓
- B) Both clients
- C) Both servers
- D) No roles

35. What is sometimes used as terminology without confusion?

- A) Client side and server side of an application ✓
- B) Peer side only
- C) Router side
- D) Network side

36. How do processes send and receive messages over the network?

- A) Through a software interface called a socket ✓
- B) Directly via hardware
- C) Using routers
- D) Via physical cables

37. What analogy is used for a process and its socket?

- A) A house and its door ✓
- B) A car and its engine
- C) A book and its pages
- D) A tree and its roots

38. When a process sends a message, what does it assume?

- A) There is a transportation infrastructure to deliver it ✓
- B) The receiver is always on
- C) No network exists
- D) Direct peer contact

39. What is a socket also referred to as?

- A) The Application Programming Interface (API) between application and network ✓
- B) A hardware port
- C) A router interface
- D) A data packet

40. What control does the application developer have on the transport-layer side of the socket?

- A) Choice of transport protocol and some parameters like buffer sizes ✓
- B) Full control over routing
- C) Hardware selection
- D) No control

Questions 41-60: Addressing Processes, Transport Services, and Internet Transport Services

41. To identify a receiving process, what information is needed?

- A) Host address and identifier for the process ✓
- B) Only hostname
- C) Only port number
- D) No identification

42. How is a host identified in the Internet?

- A) By its IP address, a 32-bit quantity ✓

- B) By its hostname only
- C) By its port number
- D) By its MAC address

43. What serves to identify the receiving socket?

- A) Destination port number ✓
- B) IP address only
- C) Hostname
- D) Buffer size

44. What port number identifies a Web server?

- A) 80 ✓
- B) 25
- C) 53
- D) 443

45. What port number is used for SMTP mail servers?

- A) 25 ✓
- B) 80
- C) 53
- D) 110

46. Where can a list of well-known port numbers be found?

- A) www.iana.org ✓
- B) Local OS files
- C) Router configs
- D) App stores

47. What is the responsibility of the transport-layer protocol?

- A) Getting messages to the socket of the receiving process ✓
- B) Routing packets
- C) Encrypting data
- D) Compressing files

48. Which transport protocols does the Internet provide?

- A) UDP and TCP ✓
- B) HTTP and SMTP
- C) DNS and FTP
- D) IP and ICMP

49. What services can a transport-layer protocol offer?

- A) Reliable data transfer, throughput, timing, and security ✓
- B) Physical transmission
- C) Routing decisions
- D) Hardware interfaces

50. What is reliable data transfer?

- A) Guaranteeing data is delivered correctly and completely ✓
- B) Allowing some data loss
- C) Only partial delivery
- D) No guarantees

51. For which applications is data loss unacceptable?

- A) E-mail, file transfer, Web documents, financial apps ✓
- B) Multimedia only
- C) Games exclusively
- D) Voice calls

52. What are loss-tolerant applications?

- A) Multimedia like audio/video that tolerate some loss ✓
- B) Financial transactions
- C) E-mail
- D) File transfers

53. What is available throughput in a communication session?

- A) The rate at which the sending process delivers bits to the receiving process ✓
- B) Fixed bandwidth only
- C) Download speed alone
- D) Upload speed alone

54. What service ensures guaranteed throughput at a specified rate?

- A) Guaranteed available throughput ✓
- B) No throughput guarantee
- C) Variable throughput
- D) Unlimited throughput

55. What are bandwidth-sensitive applications?

- A) Those with specific throughput requirements, like Internet telephony ✓

- B) Elastic apps only
- C) Text-based apps
- D) Offline apps

56. What are elastic applications?

- A) Those that use as much throughput as available, like e-mail and Web transfers ✓
- B) Fixed-rate apps
- C) Multimedia only
- D) Real-time exclusive

57. What timing guarantees might a transport protocol provide?

- A) Every bit arrives no more than a certain time later ✓
- B) No timing at all
- C) Unlimited delays
- D) Variable delays only

58. Which applications appeal to tight timing constraints?

- A) Interactive real-time like Internet telephony and games ✓
- B) File transfers
- C) E-mail
- D) Web browsing

59. What security service can a transport protocol provide?

- A) Encryption for confidentiality ✓
- B) No security
- C) Only authentication
- D) Data loss

60. What other security services are mentioned?

- A) Data integrity and end-point authentication ✓
- B) Routing security
- C) Hardware protection
- D) Firewall rules

**Questions 61-80: Transport Services by Internet,
Application-Layer Protocols, and Covered Applications**

61. What services does TCP provide?

- A) Connection-oriented and reliable data transfer ✓
- B) Unreliable only
- C) Connectionless
- D) No congestion control

62. What is TCP's connection-oriented service?

- A) Handshaking before data flow, full-duplex connection ✓
- B) No handshaking
- C) One-way only
- D) Partial duplex

63. What does TCP's reliable data transfer ensure?

- A) Data delivered without errors and in order ✓
- B) Some duplicates allowed
- C) Loss tolerated
- D) Out-of-order delivery

64. What mechanism does TCP include for the Internet's welfare?

- A) Congestion control ✓
- B) No control
- C) Bandwidth increase
- D) Delay reduction

65. What is UDP?

- A) A lightweight, connectionless protocol with minimal services ✓
- B) Reliable and connection-oriented
- C) Heavyweight
- D) With handshaking

66. Does UDP provide reliable data transfer?

- A) No, messages may not arrive or arrive out of order ✓
- B) Yes, always
- C) Only in order
- D) With guarantees

67. Does UDP include congestion control?

- A) No ✓

- B) Yes
- C) Partial
- D) Optional

68. Which services are not provided by Internet transport protocols?

- A) Throughput or timing guarantees ✓
- B) Reliable transfer
- C) Security
- D) Connection orientation

69. How can time-sensitive applications work without guarantees?

- A) They cope with the lack by design ✓
- B) They fail completely
- C) They require upgrades
- D) They use UDP only

70. Which applications use TCP?

- A) E-mail, remote access, Web, file transfer ✓
- B) Internet telephony only
- C) Games exclusively
- D) Video streaming

71. Why do Internet telephony apps often use UDP?

- A) To tolerate loss but require minimal rate ✓
- B) For reliability
- C) For congestion control
- D) For handshaking

72. What does an application-layer protocol define?

- A) Message types, syntax, semantics, and rules for sending/responding ✓
- B) Hardware interfaces
- C) Routing paths
- D) Encryption only

73. What are examples of message types in application-layer protocols?

- A) Request and response messages ✓
- B) Data packets only
- C) Error codes
- D) Acknowledgments

74. Which protocol is for the Web?

- A) HTTP ✓
- B) SMTP
- C) DNS
- D) FTP

75. Are many application-layer protocols in the public domain?

- A) Yes, specified in RFCs like HTTP ✓
- B) No, all proprietary
- C) Only some
- D) None

76. What is an example of a proprietary protocol?

- A) Skype's application-layer protocols ✓
- B) HTTP
- C) SMTP
- D) DNS

77. What distinguishes network applications from application-layer protocols?

- A) Protocols are one piece of the application ✓
- B) They are identical
- C) Applications are protocols
- D) Protocols are hardware

78. What components make up the Web application?

- A) Document standards (HTML), browsers, servers, and HTTP ✓
- B) Only servers
- C) Only browsers
- D) Routers

79. In Netflix, what is DASH?

- A) An application-level protocol for message format and sequence ✓
- B) A hardware tool
- C) A routing protocol
- D) An encryption method

80. Which applications are focused on in the chapter?

- A) Web, e-mail, DNS, video streaming, P2P ✓

- B) Games only
 - C) Databases
 - D) Spreadsheets
-

Questions 81-100: Web and HTTP Overview, Connections, and Message Format

81. What changed with the arrival of the World Wide Web in the early 1990s?

- A) It elevated the Internet to the main data network ✓
- B) It reduced usage
- C) It focused on academics only
- D) It limited access

82. What appeals most to users about the Web?

- A) It operates on demand ✓
- B) It is broadcast-only
- C) It requires scheduling
- D) It is offline

83. What features make the Web appealing?

- A) Easy publishing, hyperlinks, search, multimedia, interaction ✓
- B) High cost
- C) Limited access
- D) Text-only

84. What serves as a platform for many apps?

- A) The Web and its protocols ✓
- B) E-mail only
- C) DNS
- D) FTP

85. What is at the heart of the Web?

- A) HTTP ✓
- B) SMTP
- C) DNS
- D) TCP

86. In what RFCs is HTTP defined?

- A) RFC 1945, 7230, 7540 ✓
- B) RFC 5321
- C) RFC 1034
- D) RFC 6265

87. How do HTTP client and server communicate?

- A) By exchanging HTTP messages ✓
- B) Via UDP datagrams
- C) Through routers
- D) Directly via hardware

88. What is a Web page?

- A) Consists of objects addressable by a URL ✓
- B) A single file only
- C) Text only
- D) Images exclusively

89. What does a URL consist of?

- A) Hostname and object's path name ✓
- B) Port only
- C) IP address
- D) Protocol

90. What implements the client side of HTTP?

- A) Web browsers like Chrome ✓
- B) Servers only
- C) Routers
- D) DNS servers

91. What houses Web objects?

- A) Web servers like Apache ✓
- B) Clients
- C) Peers
- D) Mobile apps

92. How does HTTP use TCP?

- A) As its underlying transport protocol ✓

- B) It uses UDP
- C) No transport
- D) Custom protocol

93. What port does HTTP use by default?

- A) 80 ✓
- B) 25
- C) 53
- D) 443

94. Is HTTP a stateless protocol?

- A) Yes, server maintains no client info ✓
- B) No, it tracks all
- C) Partially
- D) Only for sessions

95. What is the original version of HTTP?

- A) HTTP/1.0 ✓
- B) HTTP/2
- C) HTTP/3
- D) HTTP/1.1

96. What is the majority version of HTTP in 2020?

- A) HTTP/1.1 ✓
- B) HTTP/1.0
- C) HTTP/2
- D) HTTP/3

97. In non-persistent connections, what happens for each object?

- A) A new TCP connection is established ✓
- B) Persistent use
- C) UDP used
- D) No connection

98. For a Web page with base HTML and 10 images, how many TCP connections in non-persistent?

- A) 11 ✓
- B) 1
- C) 10

- D) 2

99. What is RTT in the context of HTTP?

- A) Round-trip time for a small packet ✓
- B) Response time total
- C) Transmission time
- D) Queue time

100. Roughly, what is the total response time for non-persistent HTTP?

- A) Two RTTs plus transmission time ✓
 - B) One RTT
 - C) No RTT
 - D) Three RTTs
-

Questions 101-120: HTTP Connections, Message Format, and Cookies

101. What are shortcomings of non-persistent connections?

- A) New connection per object, TCP overhead ✓
- B) No overhead
- C) Persistent by default
- D) Low burden

102. In HTTP/1.1, what is the default mode?

- A) Persistent connections with pipelining ✓
- B) Non-persistent
- C) UDP
- D) No connections

103. In persistent connections, what happens after a response?

- A) TCP connection left open ✓
- B) Closed immediately
- C) New handshake
- D) UDP switch

104. What is pipelining in HTTP?

- A) Sending requests back-to-back without waiting ✓
- B) Waiting for each reply
- C) Closing connections
- D) Single request

105. What are the two types of HTTP messages?

- A) Request and response ✓
- B) Query and reply
- C) Send and receive
- D) Push and pull

106. In an HTTP request message, what is the first line?

- A) Request line with method, URL, version ✓
- B) Header lines
- C) Entity body
- D) Status line

107. What method is most common in HTTP requests?

- A) GET ✓
- B) POST
- C) HEAD
- D) PUT

108. What header specifies the host?

- A) Host: ✓
- B) Connection:
- C) User-agent:
- D) Accept-language:

109. What does Connection: close indicate?

- A) Non-persistent connection ✓
- B) Persistent
- C) Pipelining
- D) Keep-alive

110. What does User-agent: specify?

- A) Browser type ✓
- B) Host
- C) Language

- D) Connection

111. What is the entity body used for in GET?

- A) It is empty ✓
- B) For form data
- C) For uploads
- D) For deletions

112. When is POST used?

- A) When filling out forms ✓
- B) For debugging
- C) For uploads
- D) For deletions

113. What is HEAD similar to?

- A) GET, but for debugging ✓
- B) POST
- C) PUT
- D) DELETE

114. What does PUT allow?

- A) Uploading an object to a path ✓
- B) Deleting objects
- C) Requesting pages
- D) Forms

115. What does DELETE do?

- A) Deletes an object on the server ✓
- B) Uploads
- C) Requests
- D) Responds

116. In HTTP response, what is the first line?

- A) Status line with version, code, message ✓
- B) Headers
- C) Entity body
- D) Request line

117. What status code means OK?

- A) 200 ✓
- B) 301
- C) 400
- D) 404

118. What is 404?

- A) Not Found ✓
- B) OK
- C) Moved Permanently
- D) Bad Request

119. Why are cookies used?

- A) To identify users and track activity ✓
- B) For encryption
- C) For routing
- D) For hardware

120. How many components does cookie technology have?

- A) Four: response header, request header, cookie file, back-end database ✓
- B) Two
- C) One
- D) Three

Questions 121-140: Web Caching, HTTP/2, HTTP/3, and Electronic Mail

121. What is a Web cache also called?

- A) Proxy server ✓
- B) Origin server
- C) Client
- D) Router

122. What does a Web cache do?

- A) Satisfies HTTP requests on behalf of origin server ✓
- B) Hosts original content
- C) Routes packets
- D) Encrypts data

123. Why deploy Web caches?

- A) Reduce response time, traffic, costs ✓
- B) Increase delays
- C) Centralize data
- D) No benefits

124. What is the conditional GET?

- A) Checks if object is up to date using If-Modified-Since ✓
- B) Always gets full object
- C) Deletes cache
- D) Uploads

125. What status code for not modified?

- A) 304 ✓
- B) 200
- C) 404
- D) 301

126. What are primary goals of HTTP/2?

- A) Reduce latency, multiplexing, prioritization, compression ✓
- B) Increase connections
- C) Use UDP
- D) No changes

127. What problem does HTTP/1.1 have with single TCP?

- A) Head of Line (HOL) blocking ✓
- B) No blocking
- C) Unlimited parallel
- D) Fixed rates

128. How do browsers work around HOL in HTTP/1.1?

- A) Open multiple parallel TCP connections ✓
- B) Use single only
- C) Avoid TCP
- D) Use UDP

129. In HTTP/2, how is HOL blocking solved?

- A) Break messages into frames and interleave ✓

- B) Use multiple TCP
- C) No framing
- D) Single frame

130. What does HTTP/2 framing do?

- A) Binary encodes and interleaves frames ✓
- B) ASCII only
- C) No interleaving
- D) Large frames

131. What allows response message prioritization in HTTP/2?

- A) Assigning weights to messages ✓
- B) No prioritization
- C) Random order
- D) Fixed priority

132. What is server push in HTTP/2?

- A) Server sends multiple responses without requests ✓
- B) Client pushes
- C) No push
- D) Only one response

133. What protocol does HTTP/3 operate over?

- A) QUIC ✓
- B) TCP
- C) UDP directly
- D) HTTP/2

134. What features does QUIC provide for HTTP/3?

- A) Multiplexing, flow control, low-latency ✓
- B) No features
- C) Reliability only
- D) Congestion only

135. What are the major components of Internet mail?

- A) User agents, mail servers, SMTP ✓
- B) DNS only
- C) Web servers
- D) Routers

136. What do user agents do?

- A) Allow reading, composing, sending messages ✓
- B) Store mailboxes
- C) Route mail
- D) Encrypt

137. Where are mailboxes located?

- A) In mail servers ✓
- B) In user agents
- C) In routers
- D) In DNS

138. What is SMTP?

- A) Principal protocol for transferring mail ✓
- B) Access protocol
- C) DNS for mail
- D) HTTP for mail

139. How does SMTP restrict mail messages?

- A) Body to 7-bit ASCII ✓
- B) No restrictions
- C) Binary only
- D) Encrypted

140. Why is 7-bit ASCII a pain today?

- A) Requires encoding binary multimedia ✓
- B) No issue
- C) Supports all
- D) Faster

Questions 141-160: SMTP, Mail Formats, Access Protocols, and DNS

141. Does SMTP use intermediate mail servers?

- A) No, direct connection ✓
- B) Yes, always

- C) Optional
- D) For backups

142. What port does SMTP use?

- A) 25 ✓
- B) 80
- C) 53
- D) 110

143. Does SMTP use persistent connections?

- A) Yes ✓
- B) No
- C) UDP
- D) No connections

144. What RFC defines mail header lines?

- A) RFC 5322 ✓
- B) RFC 1945
- C) RFC 1034
- D) RFC 7540

145. Required header lines in e-mail?

- A) From: and To: ✓
- B) Subject: only
- C) Date:
- D) None

146. How do users access mailboxes on shared servers?

- A) Via HTTP or IMAP ✓
- B) Directly on host
- C) SMTP pull
- D) No access

147. Why relay through sender's mail server?

- A) For retries if recipient server down ✓
- B) No reason
- C) Direct always
- D) For encryption

148. What is IMAP?

- A) Internet Mail Access Protocol for pull ✓
- B) Push protocol
- C) SMTP variant
- D) DNS for mail

149. What does DNS provide?

- A) Hostname to IP address translation ✓
- B) Routing
- C) Encryption
- D) File transfer

150. What is DNS?

- A) Distributed database and application-layer protocol ✓
- B) Centralized server
- C) Hardware
- D) Transport protocol

151. What port does DNS use?

- A) 53 over UDP ✓
- B) 80
- C) 25
- D) 443

152. How is DNS used by other protocols?

- A) To translate hostnames to IPs ✓
- B) For data transfer
- C) For security
- D) No use

153. Why not a single centralized DNS server?

- A) Single point of failure, traffic volume, distant, maintenance ✓
- B) Scalable
- C) Low cost
- D) No issues

154. What are the classes of DNS servers?

- A) Root, TLD, authoritative ✓

- B) Local only
- C) Client servers
- D) Peers

155. What do root DNS servers provide?

- A) IP addresses of TLD servers ✓
- B) Authoritative IPs
- C) Local queries
- D) Hostnames

156. What do TLD servers handle?

- A) Top-level domains like com, org ✓
- B) Root only
- C) Authoritative
- D) Local

157. What do authoritative DNS servers do?

- A) House DNS records for organizations ✓
- B) TLD only
- C) Root
- D) Cache

158. What is a local DNS server?

- A) Acts as proxy, close to host ✓
- B) Part of hierarchy
- C) Authoritative
- D) Root

159. In DNS lookup, what type of query is from host to local DNS?

- A) Recursive ✓
- B) Iterative
- C) Both
- D) None

160. What reduces DNS query traffic?

- A) Caching ✓
- B) No reduction
- C) Centralization
- D) More servers

Questions 161-180: P2P File Distribution and Scalability

161. What is a torrent in BitTorrent?

- A) Collection of peers distributing a file ✓
- B) Single peer
- C) Server
- D) Tracker

162. What chunk size is typical in BitTorrent?

- A) 256 KBytes ✓
- B) 1 MByte
- C) 1 bit
- D) Variable

163. What infrastructure node does each torrent have?

- A) Tracker ✓
- B) Server
- C) Client
- D) Router

164. What does a new peer do in BitTorrent?

- A) Registers with tracker ✓
- B) Leaves immediately
- C) Downloads only
- D) No registration

165. How many peers does tracker send to new peer?

- A) Subset like 50 ✓
- B) All
- C) None
- D) One

166. What are neighboring peers?

- A) Those with successful TCP connections ✓
- B) All in torrent
- C) Tracker only
- D) Servers

167. What technique does BitTorrent use for requesting chunks?

- A) Rarest first ✓
- B) Common first
- C) Random
- D) Sequential

168. What algorithm does BitTorrent use for sending chunks?

- A) Trading: priority to top 4 uploaders ✓
- B) Random
- C) No priority
- D) First-come

169. What is optimistic unchoking in BitTorrent?

- A) Randomly pick one additional neighbor every 30 seconds ✓
- B) Choke all
- C) Unchoke top only
- D) No random

170. What are choked peers?

- A) Those not receiving chunks ✓
- B) Top uploaders
- C) Random
- D) All

171. In P2P vs client-server, what is self-scalability?

- A) Peers redistribute, adding capacity ✓
- B) Servers only
- C) Fixed time
- D) Linear increase

172. What is the minimum distribution time formula for client-server?

- A) $\max\{NF/us, F/d_{\min}\}$ ✓
- B) NF/us only
- C) $F/d_{\min}$ only
- D) No formula

173. For P2P, what is the minimum distribution time?

- A) $\max\{F/us, F/d_{\min}, NF/(us + \sum u_i)\}$ ✓

- B) Same as client-server
- C) Unlimited
- D) Zero

174. Why is P2P self-scaling?

- A) Distribution time bounded, not linear with N ✓
 - B) Increases with N
 - C) No peers
 - D) Servers scale
-

Questions 181-200: Video Streaming, CDNs, Socket Programming, and Summary

175. What accounts for much Internet traffic in 2020?

- A) Streaming video like Netflix, YouTube ✓
- B) E-mail
- C) Web browsing
- D) File transfer

176. In HTTP streaming, how is video sent?

- A) As ordinary file via HTTP GET ✓
- B) UDP only
- C) Custom protocol
- D) No HTTP

177. What is DASH?

- A) Dynamic Adaptive Streaming over HTTP ✓
- B) Fixed rate
- C) No adaptation
- D) UDP streaming

178. In DASH, what does the client do?

- A) Selects chunks from different bit rate versions ✓
- B) Fixed version
- C) Server selects
- D) No selection

179. What problems with single data center for video?

- A) Long paths, repeated links, single failure point ✓
- B) No issues
- C) Low cost
- D) Scalable

180. What do CDNs do?

- A) Distribute content via geographically servers ✓
- B) Centralize
- C) Use P2P only
- D) No distribution

181. What is a private CDN?

- A) Owned by content provider like Google's ✓
- B) Third-party
- C) Public
- D) Shared

182. In socket programming, what is the main task?

- A) Write client and server programs ✓
- B) Hardware code
- C) Routing
- D) No code

183. What are two types of network applications in socket programming?

- A) Open (standard protocols) and proprietary ✓
- B) Client only
- C) Server only
- D) No types

184. In the summary, what architecture is ubiquitous?

- A) Client-server ✓
- B) P2P only
- C) Centralized
- D) No architecture

185. What protocols were studied in detail?

- A) HTTP, SMTP, DNS ✓

- B) TCP only
- C) UDP
- D) IP

186. What is contrasted with client-server?

- A) P2P architecture ✓
- B) Same
- C) Hierarchical
- D) Wired

187. What is used for video distribution?

- A) CDNs ✓
- B) Single server
- C) P2P only
- D) No distribution

188. What is a protocol's definition fleshed out with?

- A) Format, order of messages, actions ✓
- B) Hardware
- C) No details
- D) Encryption

189. What service models were described?

- A) TCP and UDP ✓
- B) HTTP only
- C) DNS
- D) No models

190. What is the next chapter's focus?

- A) Transport layer details ✓
- B) Application again
- C) Physical
- D) Link

191. In email, what handles failures?

- A) Sender's server retries ✓
- B) Recipient
- C) No handling
- D) DNS

192. What is the entity body in HTTP response?

- A) The requested object ✓
- B) Headers only
- C) Status
- D) Empty

193. In DNS, what is iterative query?

- A) Reply directly to requester ✓
- B) Recursive to others
- C) No query
- D) Local only

194. In BitTorrent, what happens every 10 seconds?

- A) Recalculate top 4 peers ✓
- B) Unchoke random
- C) Leave torrent
- D) No action

195. What is the manifest file in DASH?

- A) Provides URLs and bit rates for versions ✓
- B) Video file
- C) Request
- D) Response

196. In CDNs, how is user experience improved?

- A) Direct requests to best location ✓
- B) Centralize
- C) Delay
- D) No improvement

197. In socket programming, what do processes use?

- A) Sockets for communication ✓
- B) Direct hardware
- C) No interface
- D) Routers

198. What is the black box view of DNS?

- A) Simple translation service ✓

- B) Complex routing
- C) No view
- D) Hardware

199. Why is DNS distributed?

- A) To handle scale ✓
- B) Centralized better
- C) No reason
- D) For cost

200. In P2P, why peers find each other?

- A) Capable of compatible rates tend to trade ✓
- B) Random only
- C) No finding
- D) Servers match

Wish you all the best!
Fathy.